SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT 4

Aim :

To use function types, scope, and closures, apply try-catch exception handling, and build a palindrome checker.

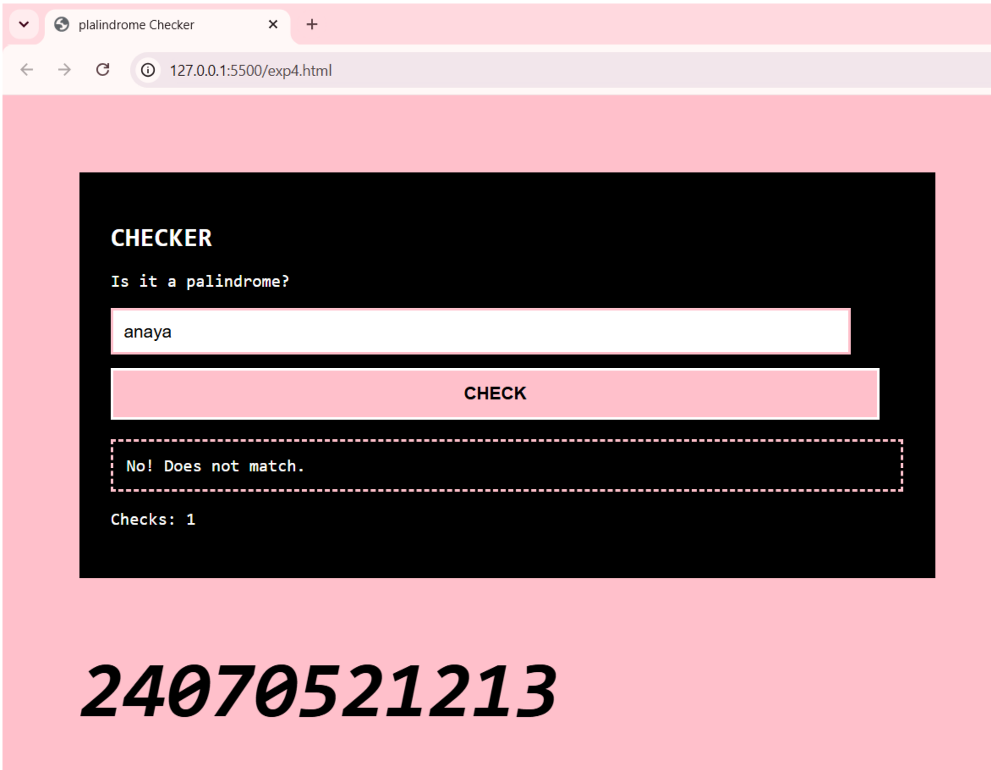
Tools Used :

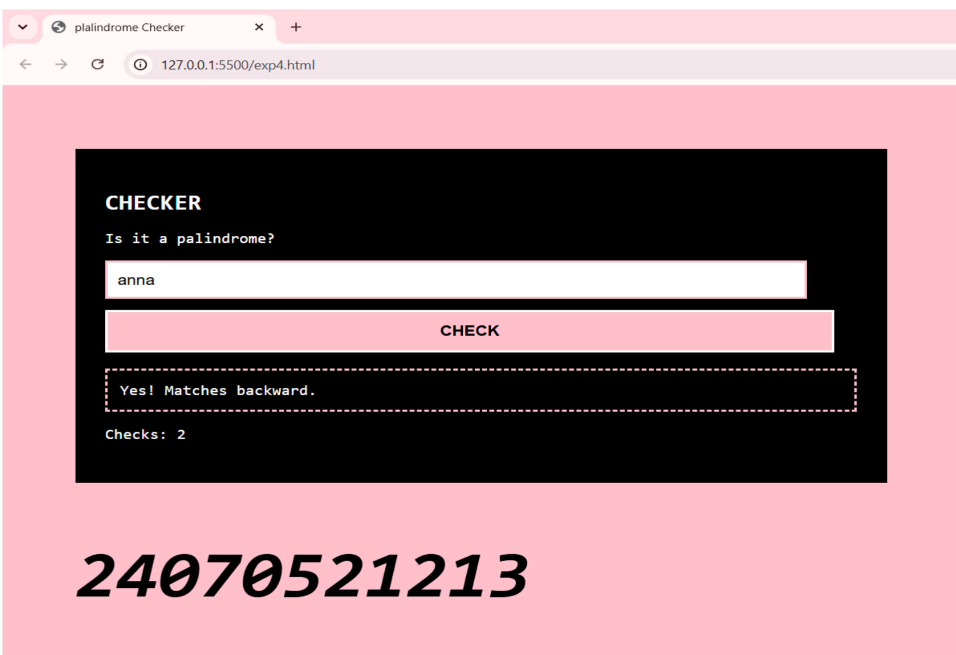
Visual Studio Code (VS Code), HTML5, CSS3, JavaScript (ES6), Google Chrome, Live Server.

Theory :

A palindrome is a word, number, or phrase that reads the same forward and backward, This experiment demonstrates the use of JavaScript function types, variable scope, closures, and exception handling. Function declarations, function expressions, and arrow functions are used to organize and reuse code efficiently. Variable scope controls the accessibility of variables, including global scope, local scope, and block scope. Closures allow an inner function to retain access to variables of its outer function even after the outer function has finished execution, making it useful for maintaining the count of palindrome checks. The `try...catch` statement is used to detect and handle runtime errors without terminating the program. The palindrome checking process involves accepting user input, converting it to lowercase, reversing the string, comparing the original string with the reversed string, displaying whether it is a palindrome, and updating the total number of checks.

OUTPUT:





Conclusion : The palindrome checker was successfully developed using HTML, CSS, and JavaScript. It correctly identifies palindrome strings while demonstrating the use of function types, variable scope, closures, and try-catch exception handling.